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(a) Determinati matricea $A = [a_{ij}]$ de dimensiune $(m+1) \times (m+1)$, $a_{ij} = (p_{m,i}, p_{m,j})$ a ecuatiilor normale relativ la baza Bernstein

$$p_{m,j} = \binom{m}{j} t^j (1-t)^{m-j}, j = \overline{0, m}, \text{ si dunctia pondere } \omega(t) = 1 \text{ pe } [0,1].$$

(b) Rezolvati sistemul de ecuatii normale pentru $m=3:3:12$, cand functia care urmeaza sa fie aproximata este $f(t)=1$. Care este solutia exacta? Afisati ,pentru fiecare m, o estimatie a numarului de conditionare, vectorul coeficientilor si eroarea asociata (modulul diferentei dintre valoarea calculata si cea exacta). Comentati rezultatul.

$$\begin{aligned} (a) \quad A_{ij} &= (p_{m,i}, p_{m,j}) = \int_{\mathbb{R}} p_{m,i} * p_{m,j} d\lambda(t) dt = \\ &= \int_0^1 \binom{m}{i} t^i (1-t)^{m-i} * \binom{m}{j} t^j (1-t)^{m-j} \omega(t) dt = \int_0^1 C_m^i t^i (1-t)^{m-i} * C_m^j t^j (1-t)^{m-j} * \\ &= C_m^i C_m^j \int_0^1 t^{i+j} * (1-t)^{2m-i-j} dt = C_m^i C_m^j * \beta(i+j, 2m-i-j) = C_m^i C_m^j * \\ &= \frac{(i+j-1)!(2m-i-j-1)!}{(2m-1)!} \end{aligned}$$

$$\text{, unde } \beta(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{(x-1)!(y-1)!}{(x+y-1)!}, \beta \text{ functia beta a lui Euler}$$

$$\begin{aligned} (b) \quad b_i &= (f, p_{m,i}) = \int_{\mathbb{R}} f(t) * p_{m,i} d\lambda(t) dt = \int_0^1 1 * \binom{m}{i} t^i (1-t)^{m-i} \omega(t) dt = C_m^i \int_0^1 t^i * \\ &= (1-t)^{m-i} * 1 dt = C_m^i \beta(i+1, m-i+1) = C_m^i * \frac{(i)(m-i)!}{(m+1)!} = \frac{m!}{i!(m-i)!} = \frac{1}{m+1} \end{aligned}$$